

ALERT meeting

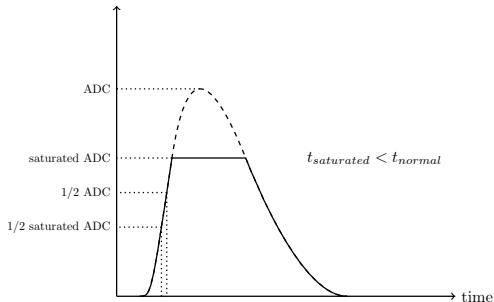
Time reconstruction of saturated signals

Felix Touchte Codjo

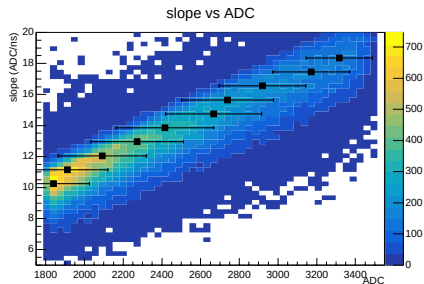
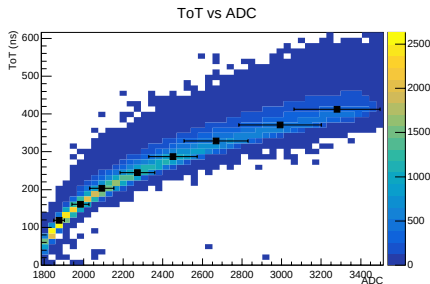
IJCLab

August 24, 2026

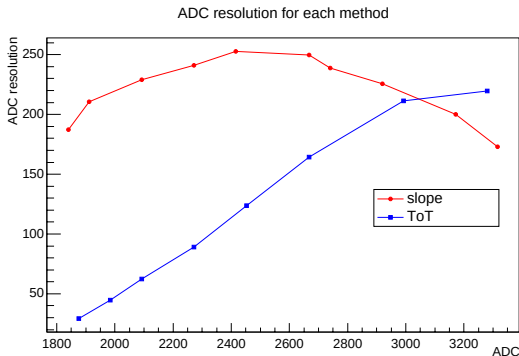
- ▶ The time of arrival is extracted at half the amplitude
- ▶ For saturated signals, this creates a bias
 - The reconstructed time is under-estimated
 - This will impact track reconstruction of α particles
 - Some hits may not even pass selection cuts and tracks can be lost
- ▶ This study aims to reconstruct a corrected time
 - The first step is to reconstruct the ADC of the saturated signals



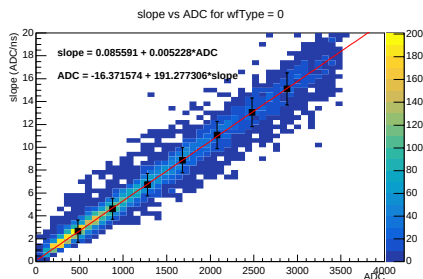
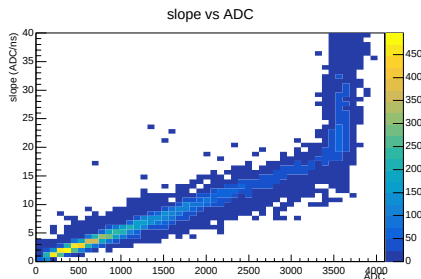
- We look at two options
 - Time over threshold (ToT) or maximum of the leading edge slope
 - We compare the two methods using non-saturated signals
 - We look at how correlated the variables are to the ADC
- The ToT as a function of the ADC is not linear
 - This might be an issue as we need to extrapolate to much larger values



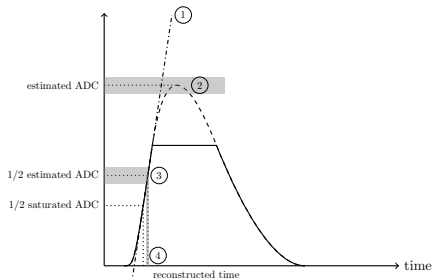
- ▶ We plot the error obtained in the previous slide
 - The slope method has a constant error
 - The ToT has a growing error
- ▶ We choose to use the maximum slope method
 - Its linear behavior will be much easier to extrapolate
 - At larger ADC, the precision will likely be better
 - Large signals are often missing their trailing edge



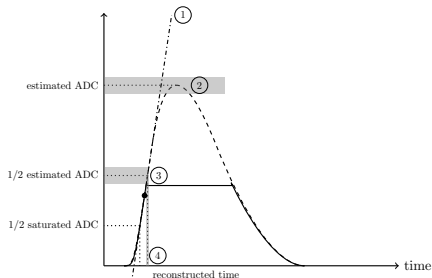
- ▶ The maximum of the slope of the signal is linearly correlated to the amplitude of the signal
 - Saturated hits seem to follow the trend
- ▶ We can use the slope to estimate the amplitude of the signal
 - We extract the ADC as a function of the slope from non-saturated signals and apply it to the saturated ones



- With the corrected amplitude, we can recompute the time
 - We use the time at half the corrected ADC
 - However, sometimes even half ADC is above the saturation threshold
 - In this case, we extrapolate the signal based on the maximum slope to determine the time at half ADC

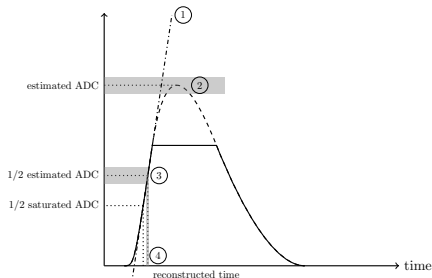


1/2 ADC below saturation
(direct interpolation)

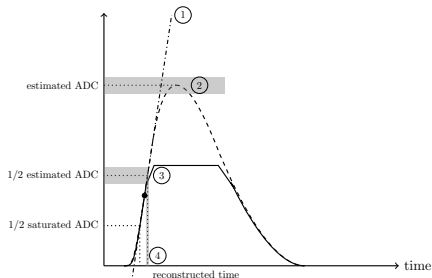


1/2 ADC below saturation
(use the maximum slope to extrapolate)

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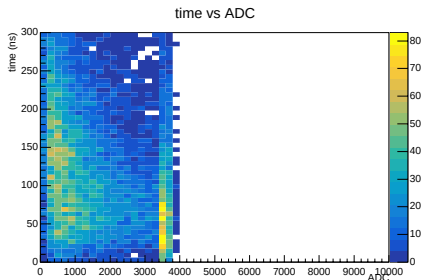


1/2 ADC below saturation
(direct interpolation)

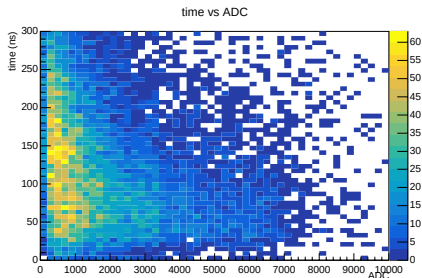


1/2 ADC below saturation
(use the previous slope to extrapolate)

- ▶ Before correction
 - We had an accumulation of hits around 3600 ADC
 - Saturated hits had abnormal tail in the low time direction
- ▶ After correction
 - We reconstruct ADCs above saturation
 - There are no more hit accumulation at low time and large amplitude
 - We move from 2244 reconstructed tracks to 2436
 - Using 10 files of run 21476, on 4He at 10.6 GeV
 - Most of the gained tracks have very high energy deposit



before



after

- ▶ The pull is ready for review [link](#)
- ▶ Other changes
 - A waveform qualified 1 cannot be assigned another type later on
 - Before some of them were requalified as 2, because of their missing trailing edge
 - The saturation limit move from 3500 to 3600 ADC
 - The upper limit of the ADC cut move from 4095 to 1e10
 - cf. [updated CCDB /calibration/alert/ahdc/raw_hit_cuts](#)